

Report of Entropy estimates based on NIST SP 800-90B non-IID track

2026-Aug-15 16:26:32.154593

1

Identification information

1.1 Identification of acquisition data from entropy source

Table 1 Identification information of acquisition data from entropy source

|                                                  |                                                                                                 |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------|
| URL of the acquisition data                      | https://github.com/usnistgov/SP800-90B_EntropyAssessment/blob/master/bin/biased-random-bits.bin |
| SHA-256 hash value of the acquisition data [hex] | 481cdac6 e2d65d45 656c2123 4125eaf2 6df18a49 037f15ff d40002b3 5e547586                         |
| Last write time                                  | 2021-Sep-20 10:22:43                                                                            |

- Name of the submitter of the acquisition data :
- Brief explanation of the acquisition data (or entropy source) :

1.2 Identification of analysis environment

Table 2 Identification information of analysis environment

|                      |                        |                                                        |
|----------------------|------------------------|--------------------------------------------------------|
| Analysis tool        | Name                   | Another entropy estimation tool with extensions        |
|                      | Versioning information | 1.1.10                                                 |
|                      | built as               | 64-bit application                                     |
|                      | built by               | Intel C++ Compiler ( __INTEL_LLVM_COMPILER: 20250303 ) |
|                      | linked libraries       | Boost C++ 1.92.0                                       |
| Analysis environment | Hostname               | ██████████                                             |
|                      | CPU information        | Intel(R) Core(TM) i5-██████████                        |
|                      | Physical memory size   | ██████ MiB                                             |
|                      | OS name                | Microsoft Windows 11 Pro                               |
|                      | OS version             | 10.0.26200 N/A Build 26200                             |
|                      | System type            | 64-bit                                                 |
|                      | Username               | ██████                                                 |

1.3 Identification of analysis conditions

Table 3 Identification information of analysis conditions

|                                                             |         |
|-------------------------------------------------------------|---------|
| Number of samples                                           | 1000000 |
| Bits per sample                                             | 1       |
| Use Longest Common Prefix <sup>*1</sup> for 6.3.5 and 6.3.6 | True    |

1.4 Identification of analysis method

NIST SP 800-90B [1] 6.3 with corrections [2] is applied

<sup>\*1</sup> See [3] and [4]

**2**

**Executive summary**

2.1 Numerical results of min-entropy estimates based on non-IID track

Table 4 Numerical results

| Estimator                                                                                   | $H_{\text{bitstring}}^{\text{a}}$<br>[bit / 1 - bit] | Notes to $H_{\text{bitstring}}$ |
|---------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------|
| The Most Common Value Estimate                                                              | 0.0286331                                            | see 3.1                         |
| The Collision Estimate                                                                      | 0.0285138                                            | see 3.2                         |
| The Markov Estimate                                                                         | 0.029123                                             | see 3.3                         |
| The Compression Estimate                                                                    | 0.0177666                                            | see 3.4                         |
| The t-Tuple Estimate                                                                        | 0.0264893                                            | see 3.5                         |
| The Longest Repeated Substring (LRS) Estimate                                               | 0.0558814                                            | see 3.6                         |
| Multi Most Common in Window Prediction Estimate                                             | 0.0286349                                            | see 3.7                         |
| The Lag Prediction Estimate                                                                 | 0.0405998                                            | see 3.8                         |
| The MultiMMC Prediction Estimate                                                            | 0.0286346                                            | see 3.9                         |
| The LZ78Y Prediction Estimate                                                               | 0.028635                                             | see 3.10                        |
| The intial entropy source estimate [bit / 1 -bit]<br>$H_I = H_{\text{bitstring}}$           | 0.0177666                                            |                                 |
| <sup>a</sup> Entropy estimate of the sequential dataset [source: NIST SP 800-90B [1] 3.1.3] |                                                      |                                 |

2.2 Visual comparison of min-entropy estimates from binary samples

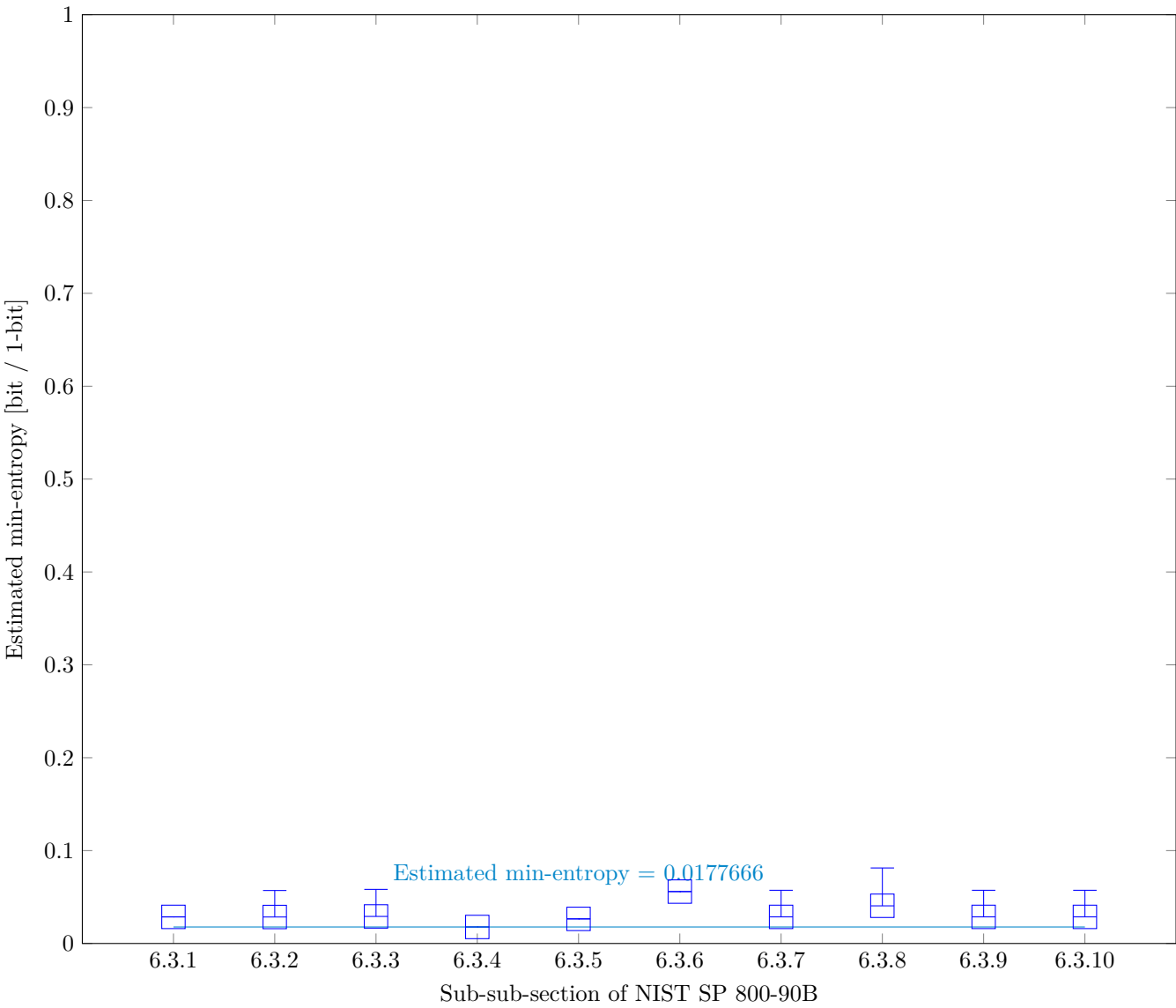


Fig. 1 Estimated Min-Entropy using §6.3 of NIST SP 800-90B

3

 Detailed results of analysis from original samples

3.1 The Most Common Value Estimate (NIST SP 800-90B Section 6.3.1)

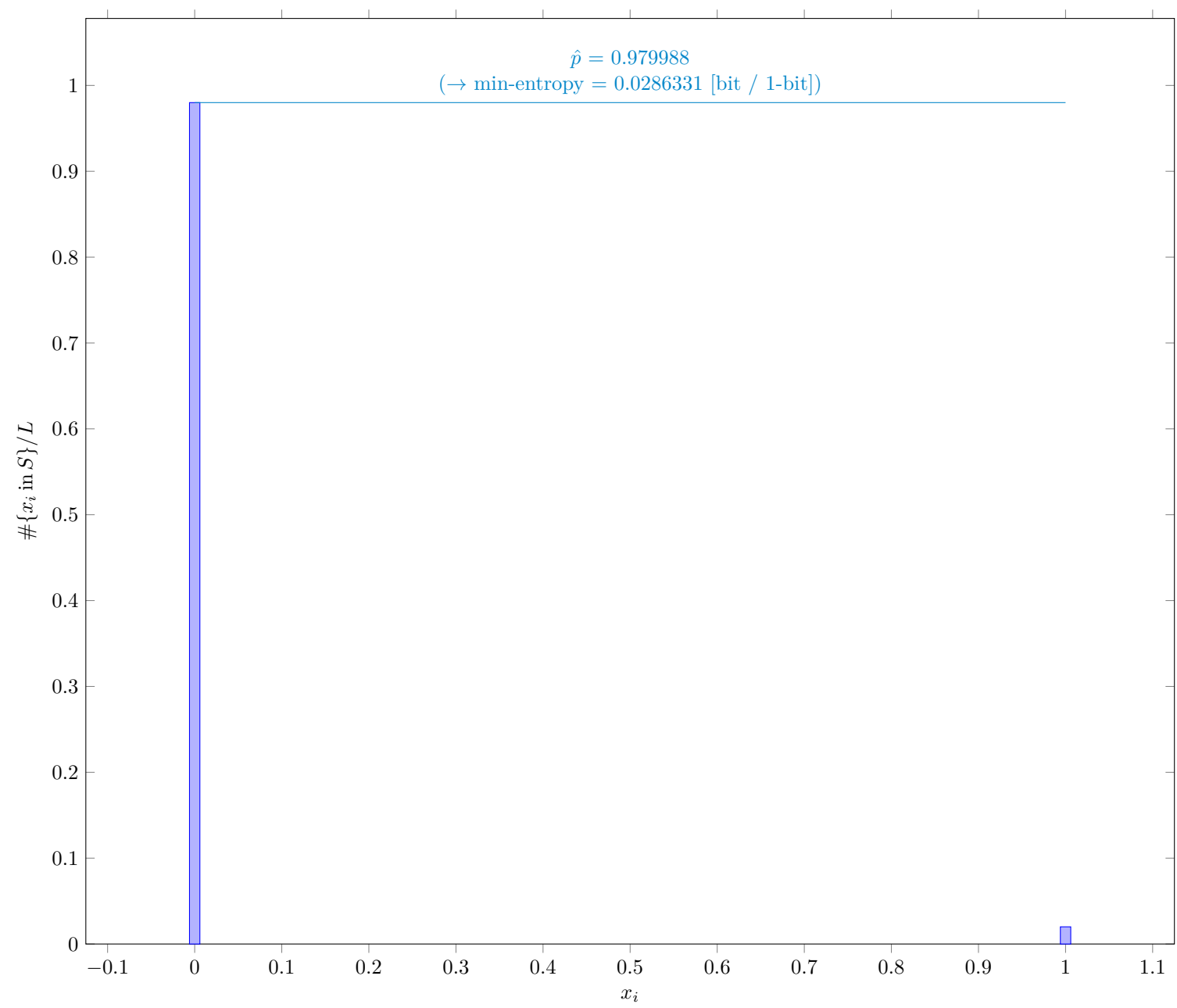


Fig. 2 Distribution of  $x_i$

3.1.1 Supplemental information for traceability

Table 5 Supplemental information for traceability (NIST SP 800-90B Section 6.3.1)

| Symbol    | Value    |
|-----------|----------|
| mode      | 979988   |
| $\hat{p}$ | 0.979988 |
| $p_u$     | 0.980349 |

3.2 The Collision Estimate (NIST SP 800-90B Section 6.3.2)

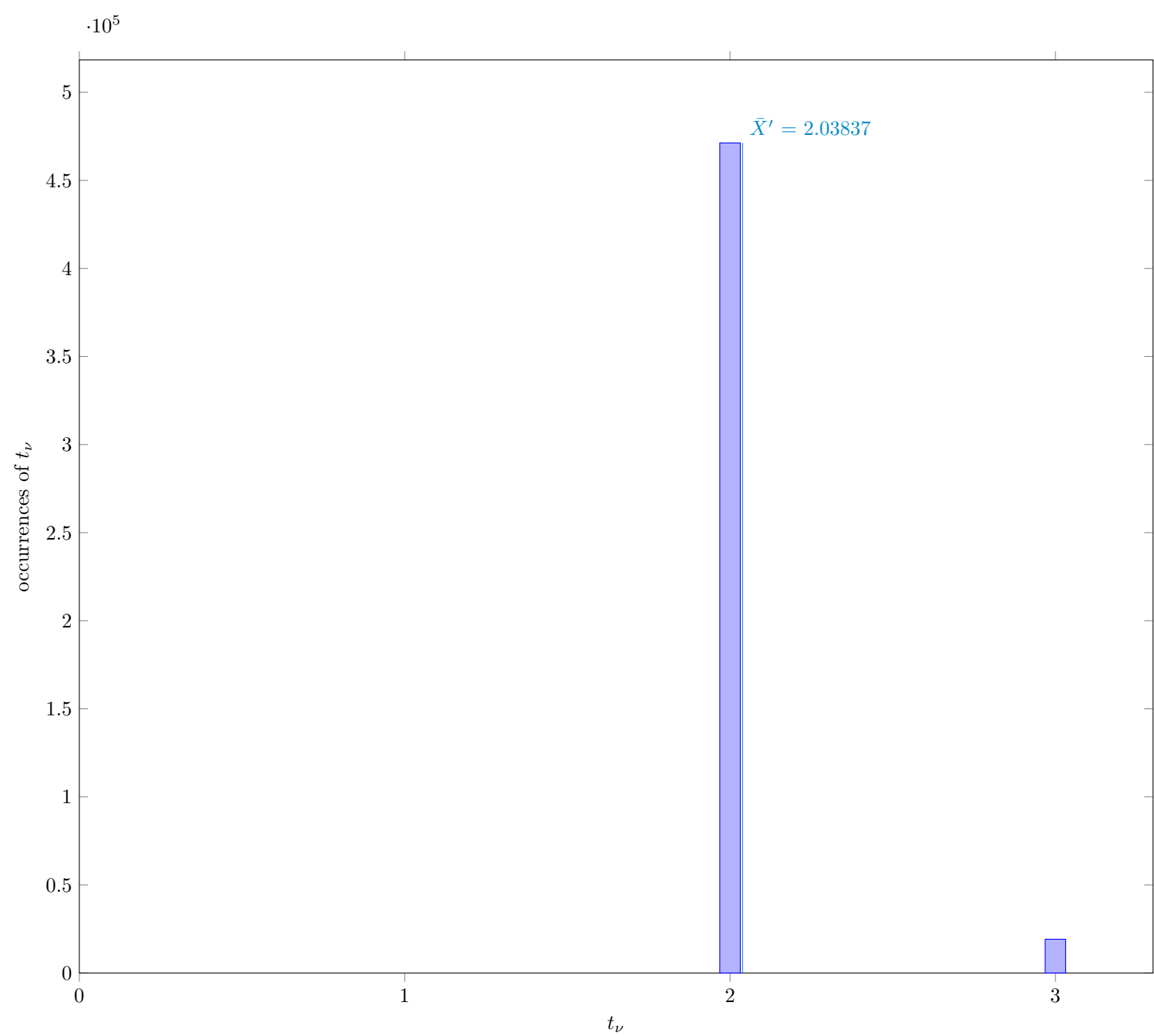


Fig. 3 Distribution of intermediate value  $t_\nu$

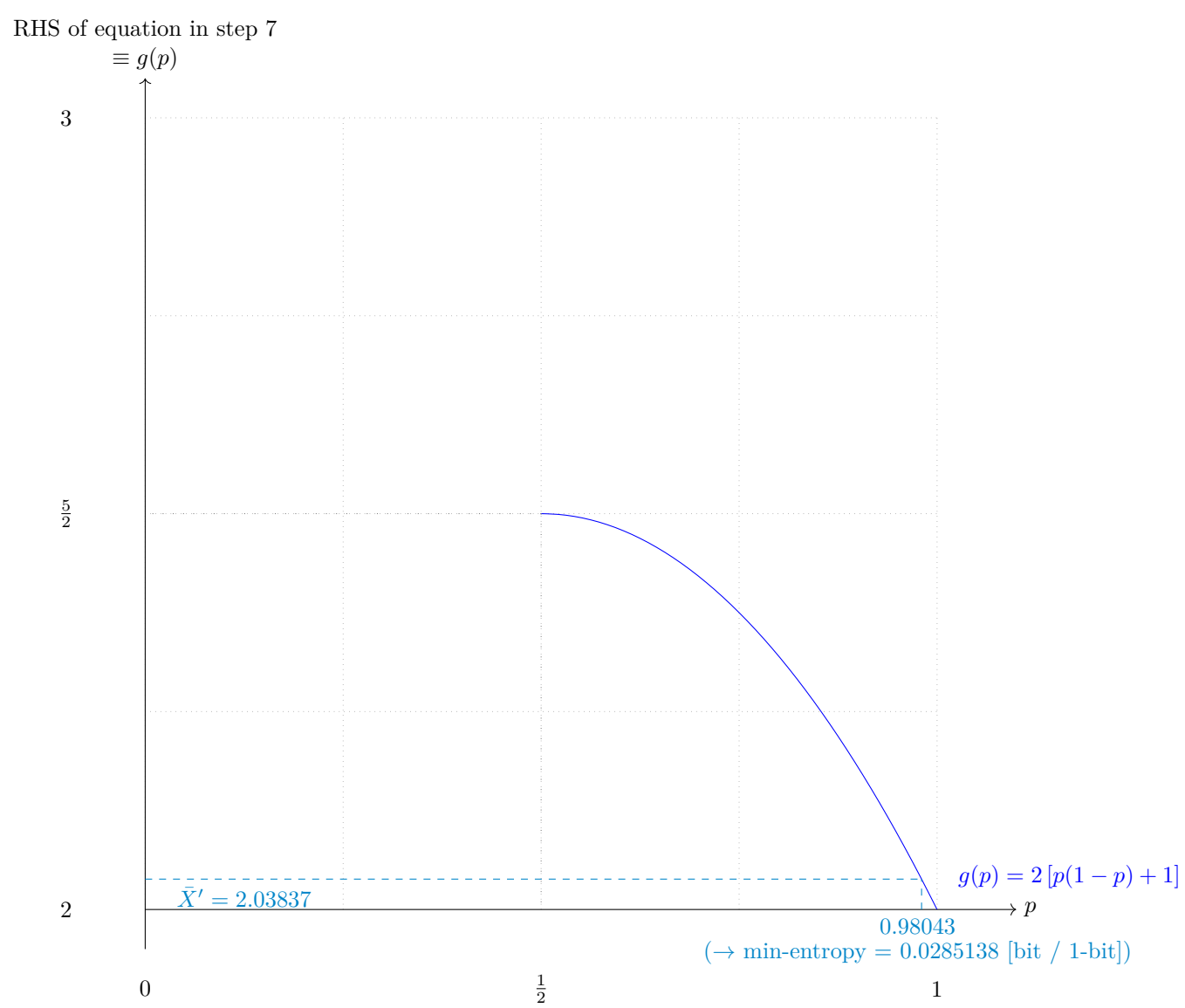


Fig. 4 Solution to the equation in step 7



3.2.1 Supplemental information for traceability

Table 6 Supplemental information for traceability (NIST SP 800-90B Section 6.3.2)

| Symbol         | Value    |
|----------------|----------|
| $p$            | 0.98043  |
| $\bar{X}$      | 2.03909  |
| $\bar{X}'$     | 2.03837  |
| $\hat{\sigma}$ | 0.193803 |

## 3.3 The Markov Estimate (NIST SP 800-90B Section 6.3.3)

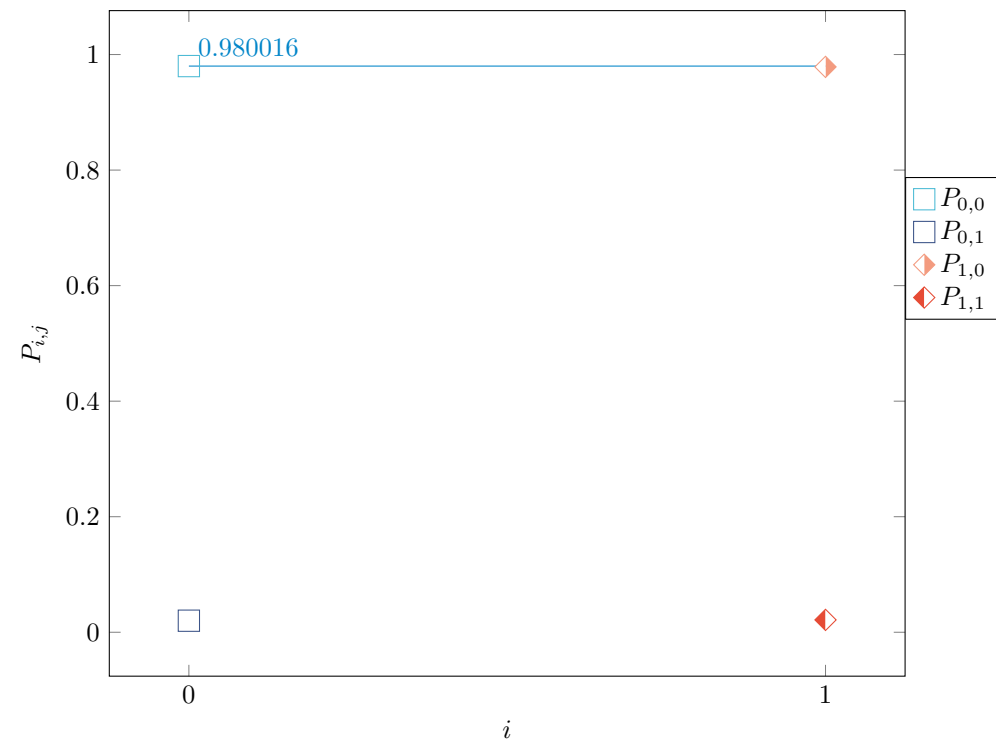
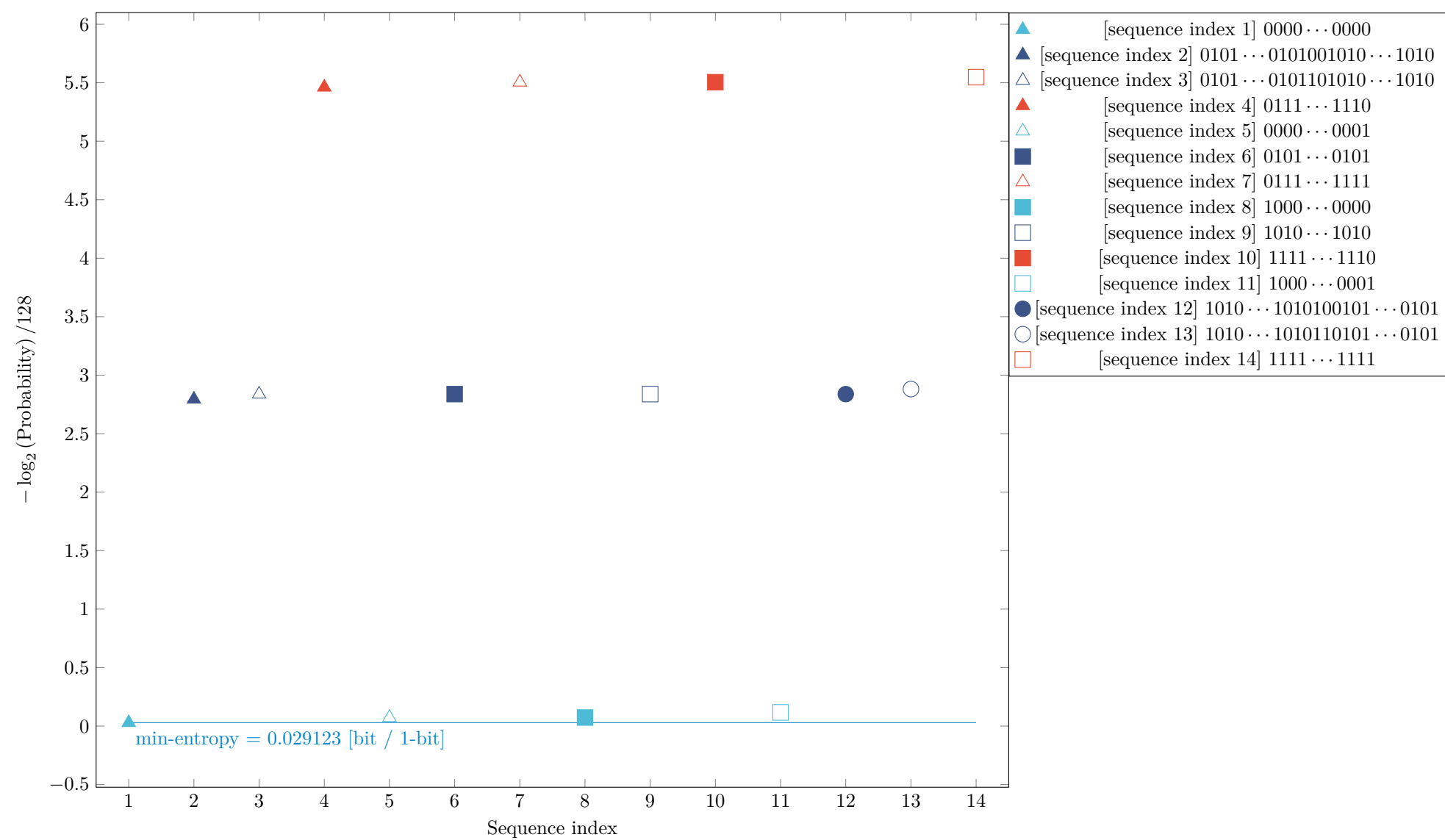
Fig. 5 Transition probability  $P_{i,j}$  of §6.3.3 of NIST SP 800-90B

Fig. 6 Estimated Min-Entropy using §6.3.3 of NIST SP 800-90B



3.4 The Compression Estimate (NIST SP 800-90B Section 6.3.4)

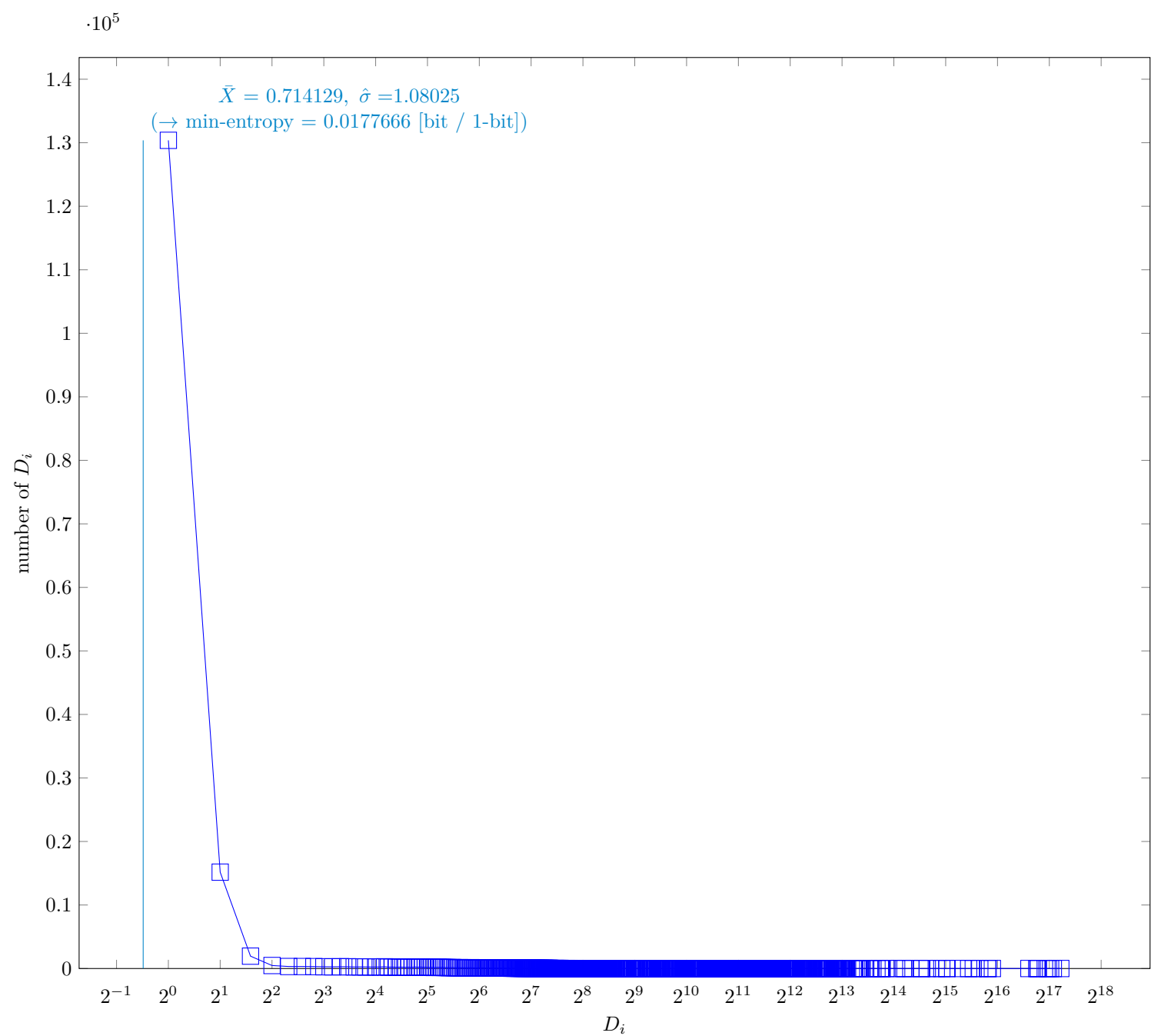


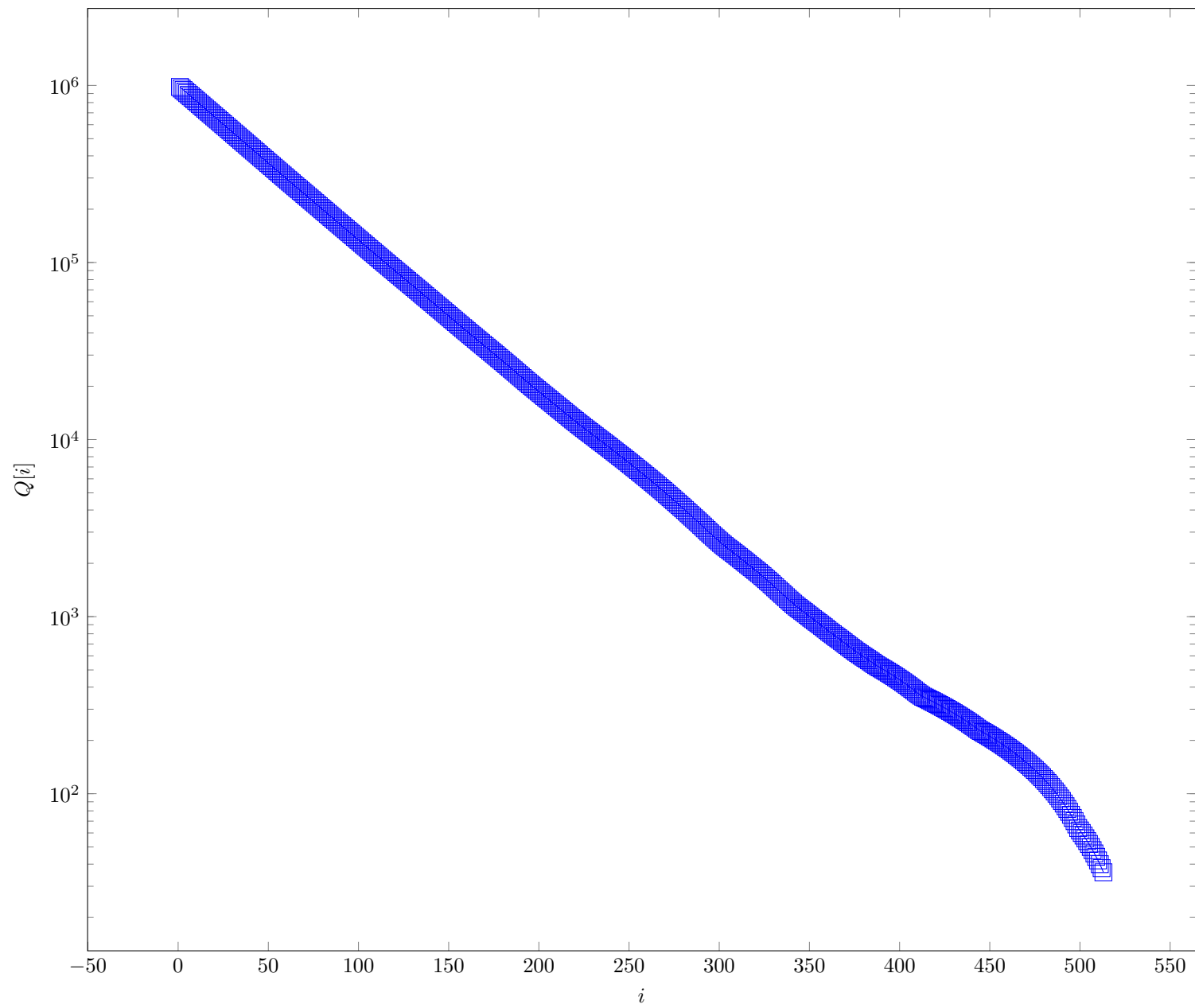
Fig. 7 Distribution of intermediate value  $D_i$

3.4.1 Supplemental information for traceability

Table 7 Supplemental information for traceability (NIST SP 800-90B Section 6.3.4)

| Symbol         | Value    |
|----------------|----------|
| $p$            | 0.928775 |
| $\bar{X}$      | 0.714129 |
| $\hat{\sigma}$ | 1.08025  |
| $\bar{X}'$     | 0.707292 |

## 3.5 The t-tuple Estimate (NIST SP 800-90B Section 6.3.5)

Fig. 8 Intermediate value  $Q[i]$  in §6.3.5 of NIST SP 800-90B

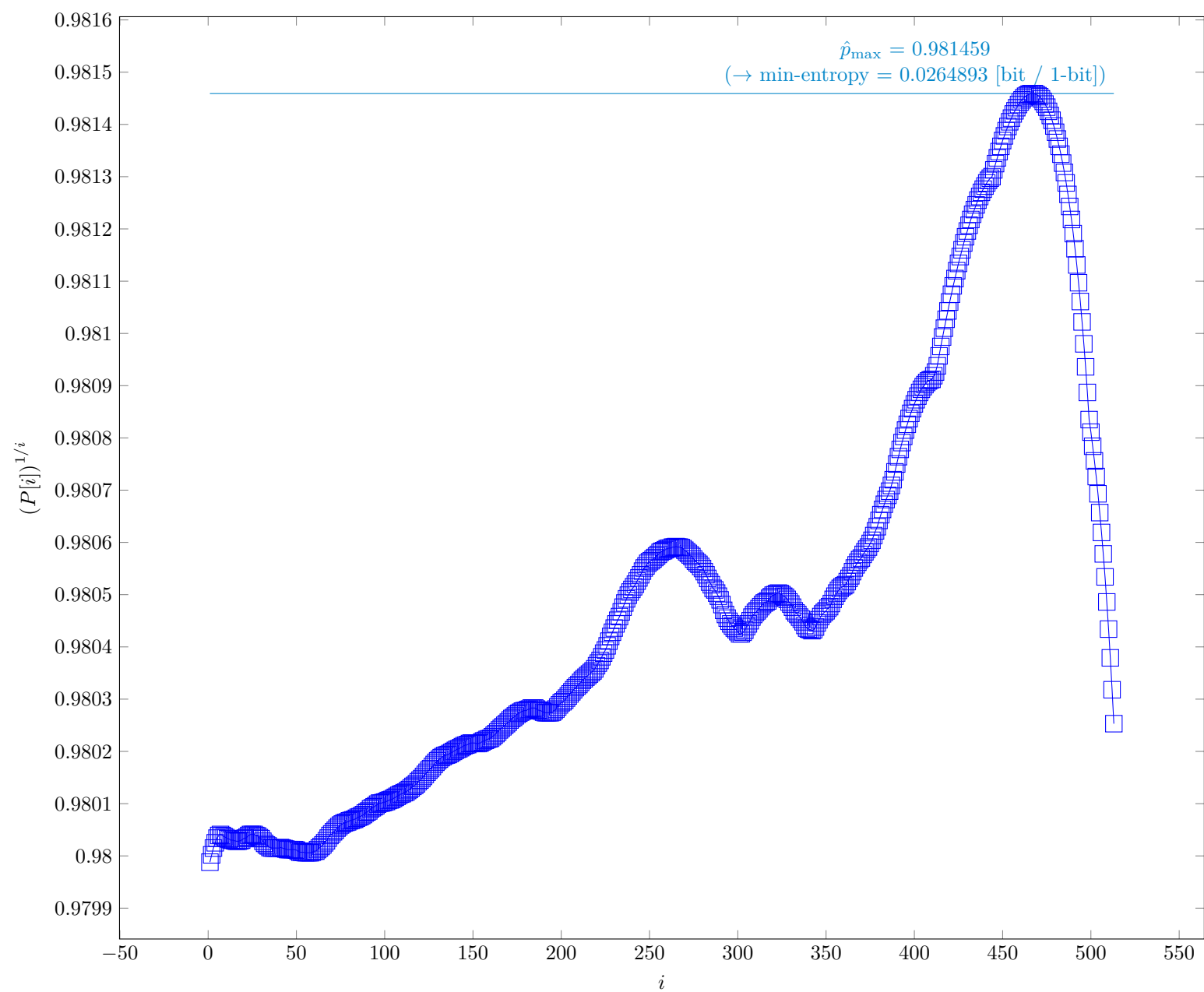


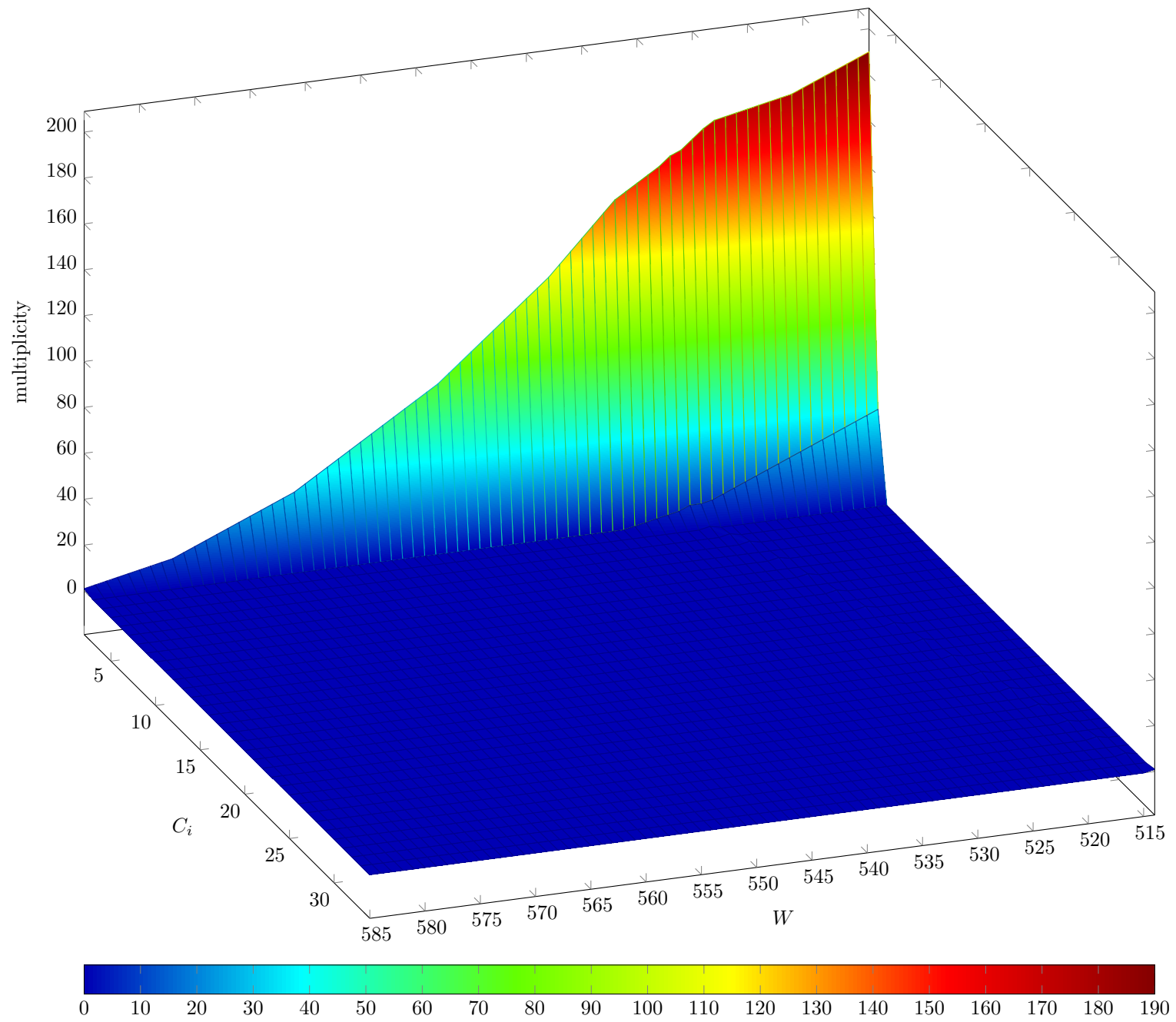
Fig. 9  $P[i]^{1/i}$  in §6.3.5 of NIST SP 800-90B

3.5.1 Supplemental information for traceability

Table 8 Supplemental information for traceability (NIST SP 800-90B Section 6.3.5)

| Symbol           | Value    |
|------------------|----------|
| $t$              | 513      |
| $\hat{p}_{\max}$ | 0.981459 |
| $p_u$            | 0.981807 |

## 3.6 The LRS Estimate (NIST SP 800-90B Section 6.3.6)

Fig. 10 Distribution of the multiplicity of  $W$ -tuple having the same  $C_i$  value in Step 3 of §6.3.6 of NIST SP 800-90B

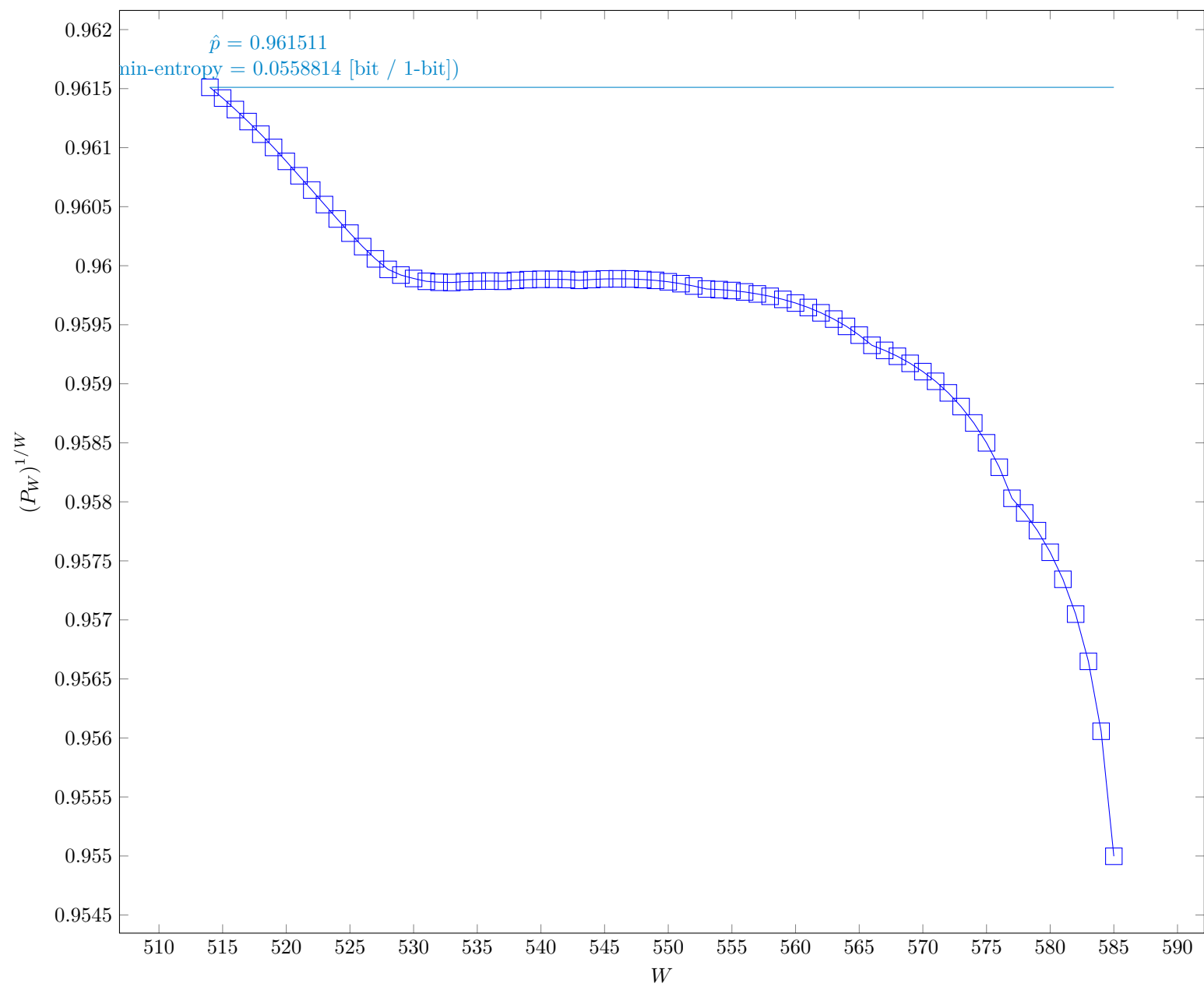


Fig. 11 Estimated average collision probability per string symbol in Step 3 of §6.3.6 of NIST SP 800-90B



3.6.1 Supplemental information for traceability

Table 9 Supplemental information for traceability (NIST SP 800-90B Section 6.3.6)

| Symbol    | Value    |
|-----------|----------|
| $u$       | 514      |
| $v$       | 585      |
| $\hat{p}$ | 0.961511 |
| $p_u$     | 0.962007 |

3.7 Multi Most Common in Window Prediction Estimate (NIST SP 800-90B Section 6.3.7)

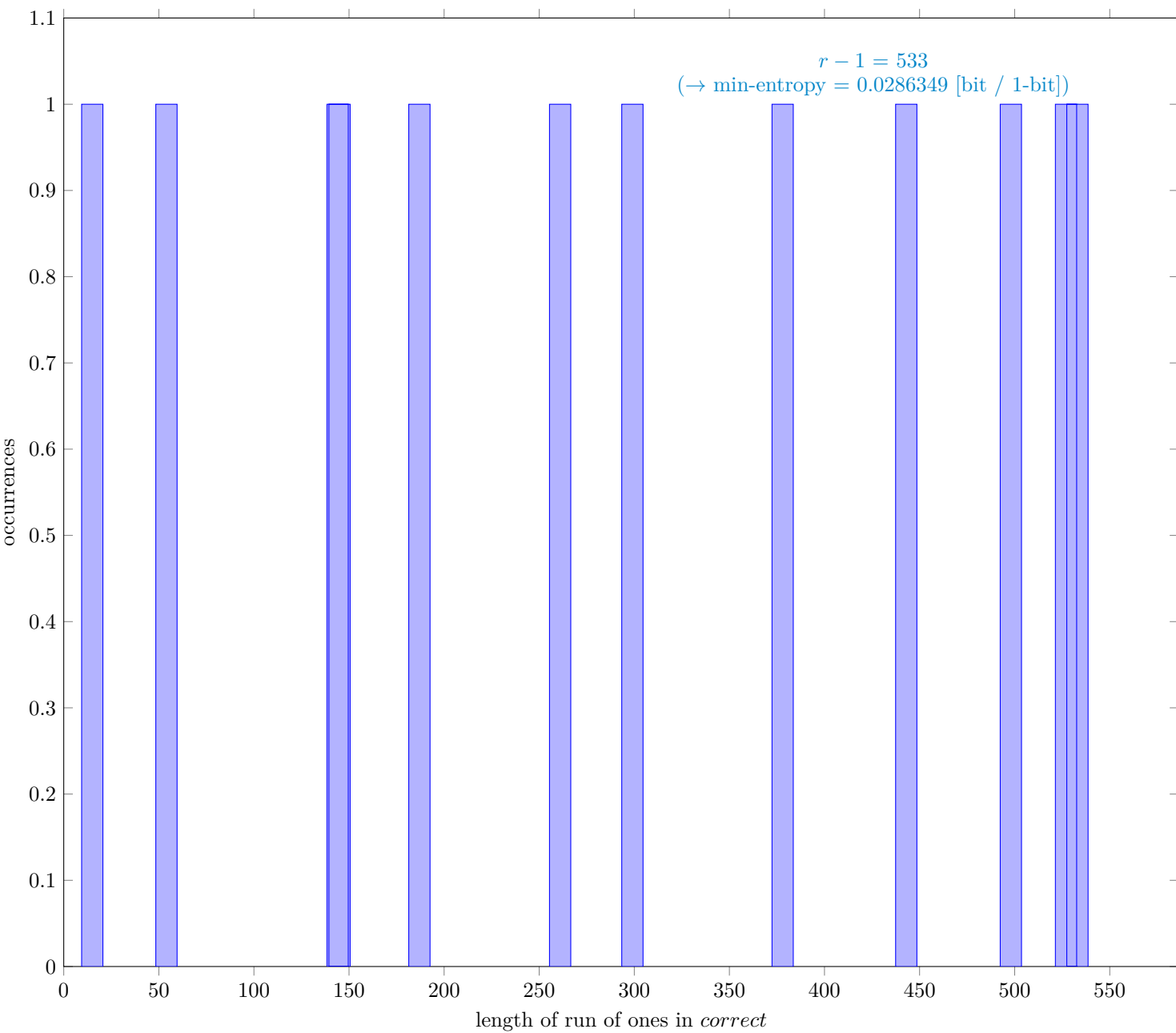


Fig. 12 Distribution of *correct*

3.7.1 Supplemental information for traceability

Table 10 Supplemental information for traceability (NIST SP 800-90B Section 6.3.7)

| Symbol               | Value    |
|----------------------|----------|
| $N$                  | 999937   |
| $C$                  | 979925   |
| $P_{\text{global}}$  | 0.979987 |
| $P'_{\text{global}}$ | 0.980347 |
| $r$                  | 534      |
| $P_{\text{local}}$   | 0.972635 |



3.8 Lag Prediction Estimate (NIST SP 800-90B Section 6.3.8)

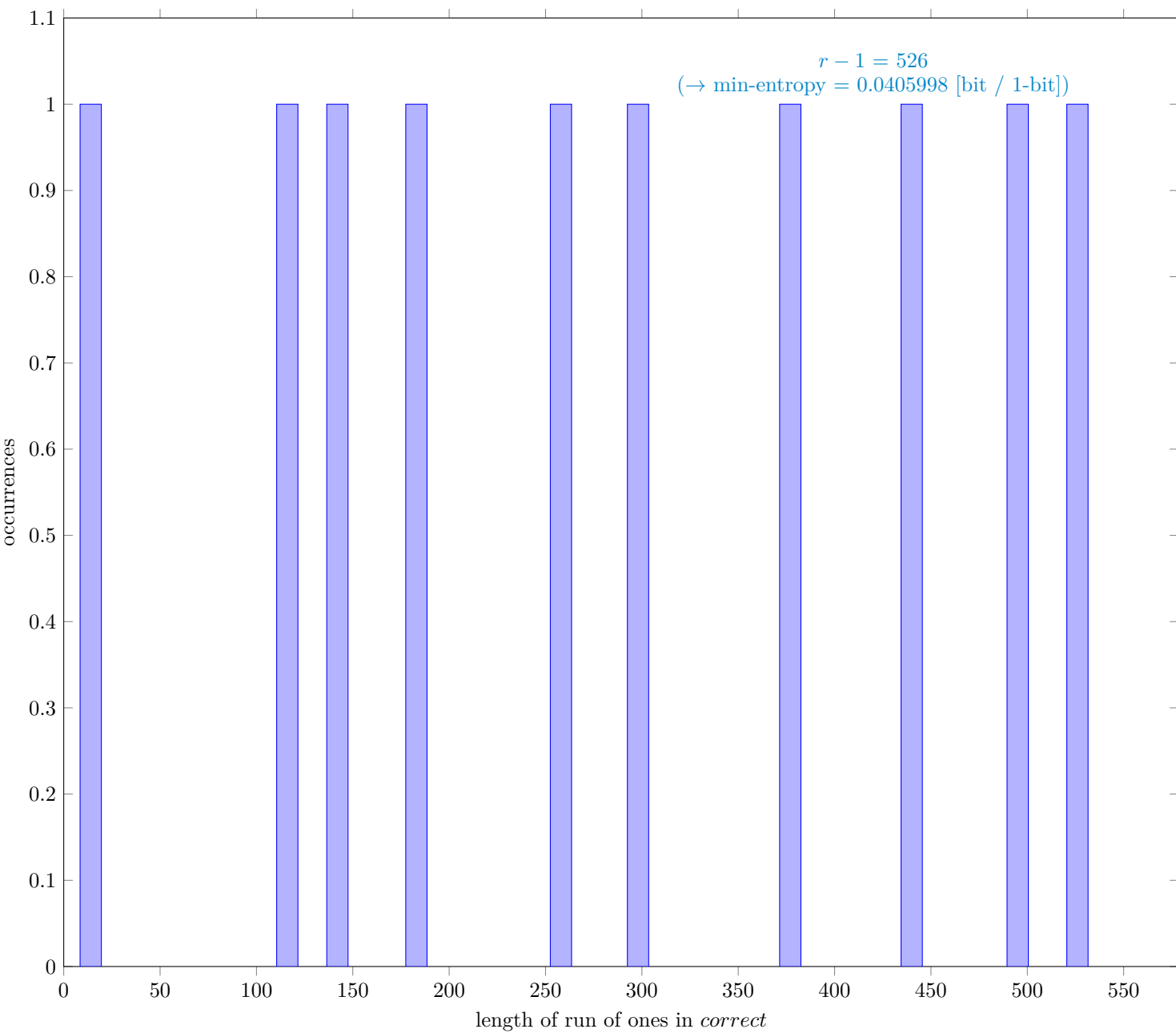


Fig. 13 Distribution of *correct*

3.8.1 Supplemental information for traceability

Table 11 Supplemental information for traceability (NIST SP 800-90B Section 6.3.8)

| Symbol               | Value    |
|----------------------|----------|
| $N$                  | 999999   |
| $C$                  | 959838   |
| $P_{\text{global}}$  | 0.959839 |
| $P'_{\text{global}}$ | 0.960345 |
| $r$                  | 527      |
| $P_{\text{local}}$   | 0.972251 |

3.9 The MultiMMC Prediction Estimate (NIST SP 800-90B Section 6.3.9)

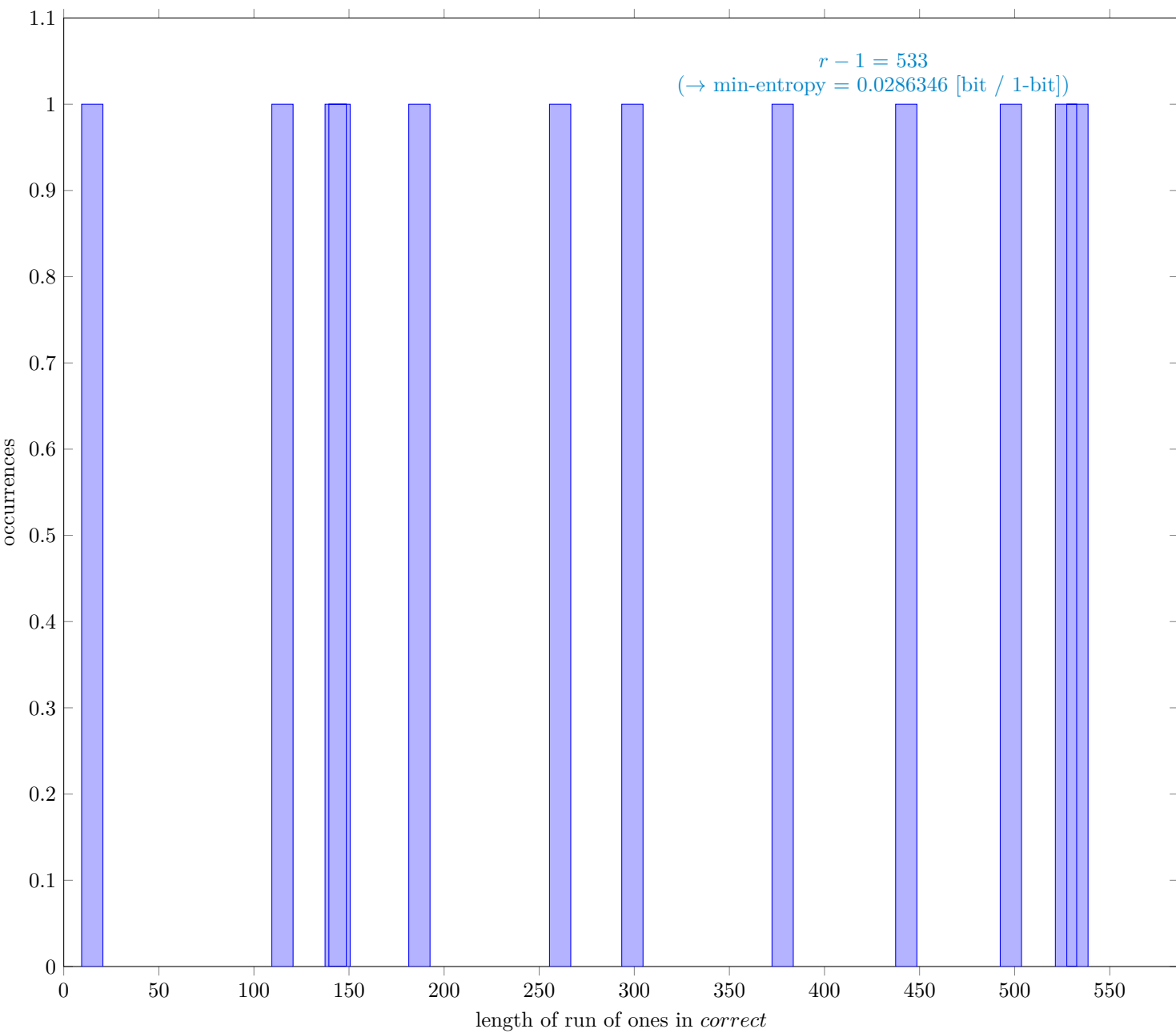


Fig. 14 Distribution of *correct*

3.9.1 Supplemental information for traceability

Table 12 Supplemental information for traceability (NIST SP 800-90B Section 6.3.9)

| Symbol               | Value    |
|----------------------|----------|
| $N$                  | 999998   |
| $C$                  | 979985   |
| $P_{\text{global}}$  | 0.979987 |
| $P'_{\text{global}}$ | 0.980348 |
| $r$                  | 534      |
| $P_{\text{local}}$   | 0.972635 |

### 3.10 The LZ78Y Prediction Estimate (NIST SP 800-90B Section 6.3.10)

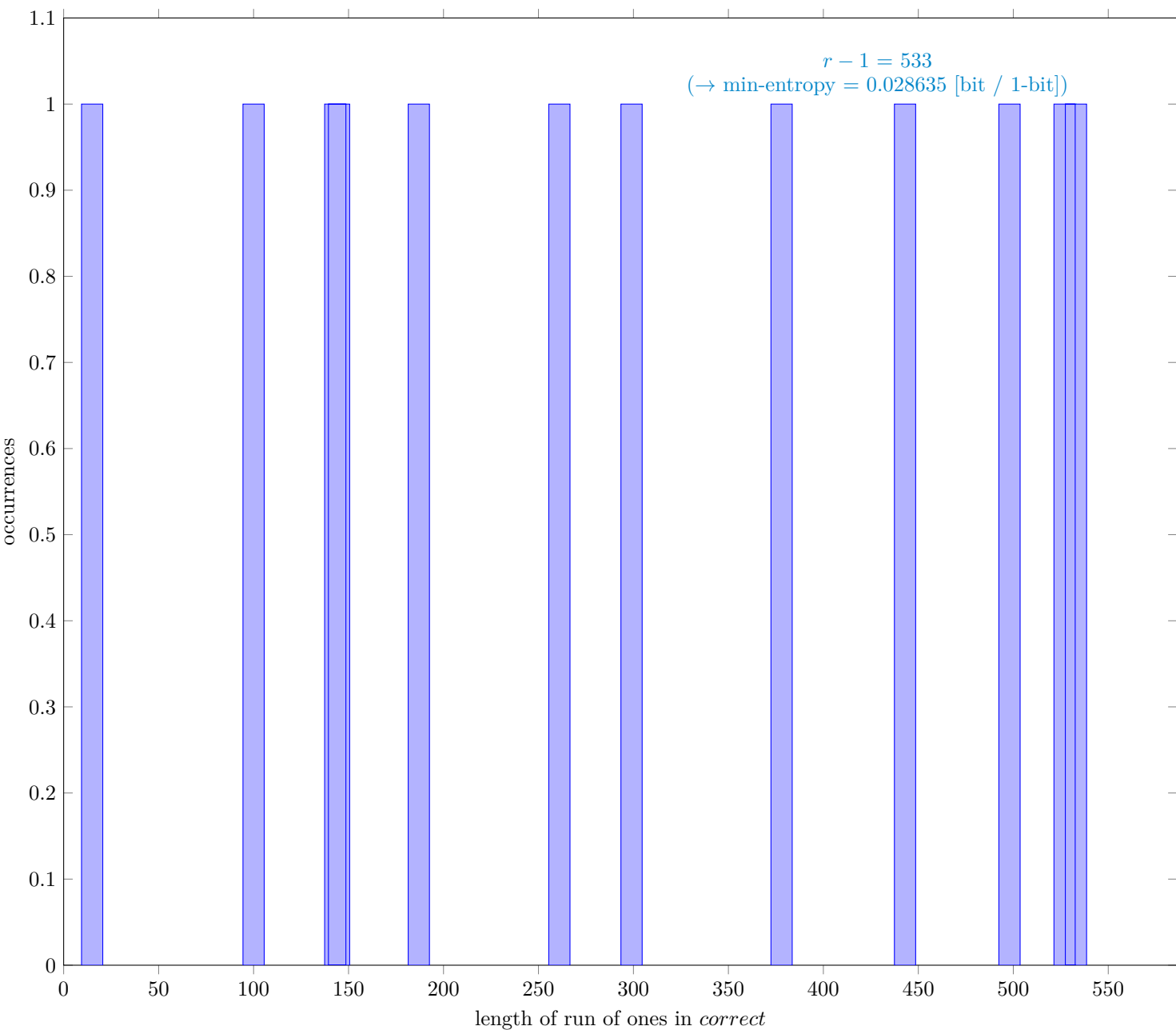


Fig. 15 Distribution of *correct*

#### 3.10.1 Supplemental information for traceability

Table 13 Supplemental information for traceability (NIST SP 800-90B Section 6.3.10)

| Symbol               | Value    |
|----------------------|----------|
| $N$                  | 999983   |
| $C$                  | 979970   |
| $P_{\text{global}}$  | 0.979987 |
| $P'_{\text{global}}$ | 0.980347 |
| $r$                  | 534      |
| $P_{\text{local}}$   | 0.972635 |

## 4 References

- [1] Meltem Sönmez Turan, Elaine Barker, John Kelsey, Kerry A. McKay, Mary L. Baish, Mike Boyle, *Recommendation for the Entropy Sources Used for Random Bit Generation*, NIST Special Publication 800-90B, Jan. 2018 <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-90B.pdf>
- [2] G. Sakurai, *Proposed list of corrections for NIST SP 800-90B 6.3 Estimators*, June 2025 [https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/ProposedListOfCorrections\\_SP800-90B.pdf](https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/ProposedListOfCorrections_SP800-90B.pdf)
- [3] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, *Introduction to Algorithms (fourth edition)*, The MIT Press. <https://mitpress.mit.edu/9780262046305/introduction-to-algorithms/>
- [4] G. Sakurai, *Implementation Notes for entropy estimation based on NIST SP800-90B non-IID track*, May 2026 [https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/SP800-90B\\_EntropyEstimate\\_ImplementationNotes.pdf](https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/SP800-90B_EntropyEstimate_ImplementationNotes.pdf)